
Mercedes-Benz
Daimler AG

Adaption Module as Vehicle Control ADM3



Control Unit Operating Manual

Revision 11.01

Version History

Date	Revision	Changes
2013-12-11	11.01	Chapter 11.5.13. (ERC1 – Electronic Retarder Controller 1 (receive)): Meaning of Byte 7 in SAE J1939-71 message ERC1 corrected
2013-03-21	11.00	- Chapter 3.3. (Functional description of the ADM3 pins): Parameter description adapted to changes in parameter values for the 4 relays, the Wait to Start Lamp and the digital input Pin DSF3 - Chapter 4.1.1. and 4.2.1. (Group 01 – CAN Configuration): Parameter 01/20 (SAE J1939 EC1 Cycle Time) added - Chapter 4.1.2. and 4.2.2. (Group 02 – Veh Par I / Relay 2): Value 9 (Vehicle Overspeed) added for parameter 02/03 (Configuration Relay 2 / Grid heater), parameter 02/16 (Configuration Wait to Start Lamp (PIN 21/7)) added - Chapter 4.1.8. and 4.2.8. (Group 08 – Vehicle Speed Sensor): Default value of parameter 08/12 (Maximum valid C3-Voltage (Low-Level)) changed to 4V - Chapter 4.1.13. and 4.2.13. (Group 13 – Inputs): Value 3 (Quit Signal for CC with Stalk Switch) of parameter 13/38 (Input DSF3) added, parameter 13/40 (Configuration Engine Shutdown Override) added - Chapter 4.1.14. and 4.2.14. (Group 14 – Relay 3/4, AdBlue, NOx, misc.): Value 14 (Vehicle Overspeed) added for parameter 14/01 (Configuration Relay 3 (IWK3)), value 15 (Vehicle Overspeed) added for parameter 14/10 (Configuration Relay 4 (IWK4)) - Chapter 4.1.16. and 4.2.16. (Group 16 – Relay 1 / Starter Lockout): Value 10 (Vehicle Overspeed) added for parameter 16/01 (Output Relay 1 / Starter Lockout) - Chapter 7.1.2.2. (Cruise control when using a stalk switch): Fig. 11: Stalk switch wiring adapted, because QUIT signal can be connected to DSF2 or DSF3 now, Pin 12/07 (DSF2) extended by alternative wiring at Pin 12/08 (DSF3), reference to parameter 13/38 (Input DSF3) value 3 (Quit Signal for CC with Stalk Switch) added as alternative to respective value for DSF2 - Chapter 7.6.12. (Vehicle Overspeed Lamp (ECE-R89)) added - Chapter 7.11.1. (Engine Protection Shutdown): The Engine Shutdown Override functionality is adapted to the extended inputs hardwired and J1939 CAN signal in message CCVS1 or ESS - Chapter 9.1. (Fault codes diagnosis version 210) and 9.2. (Fault codes listed by K-line code): Fault codes SPN/FMI 1237/09 and 1237/19 added, SPN for faults Reductant Temperature Sensor Circuit High Input and Circuit Low Input changed from 3515 to 4337 (only for Diesel engines) - Chapter 11.4.1. (Miscellaneous messages) and 11.5.36. (CCSS – Cruise Control/Vehicle Speed Setup): J1939 message CCSS – Cruise Control/Vehicle Speed Setup added - Chapter 11.4.1. (Miscellaneous messages) and 11.5.48. (PropB_0A – Proprietary B_0A): Proprietary J1939 message PropB_0A – Proprietary B_0A added - Chapter 11.5.19. (EEC2 – Electronic Engine Controller 2 (send)): Signal Actual Maximum Available Engine – Percent Torque in J1939 message EEC2 (send) in Byte 7 changed to supported - Chapter 11.5.27. (AUXIO1 – Auxiliary Input/Output Status 1): Signal Vehicle Overspeed in J1939 message AUXIO1 in Byte 4, Bits 6,5 added - Chapter 11.5.40. (CCVS1 – Cruise Control/Vehicle Speed 1 (send)): Byte 1, Bits 6,5 (Cruise Control Pause Switch) changed to supported
2013-03-21	10.00c	- Chapter 3.2. (Wiring diagram of ADM3): Pin 12/7 is connected to a switch to ground, description of pin 12/8 ST_SP is replaced by DSF3 - Chapter 3.3.3. (12 pin Connectors): shortcut for input pin 12/08 MOKL is replaced by DSF3, “engine hood bus” ist replaced by “engine crank inhibit” - Chapter 3.4.3. (Digital inputs (DE)): Technical data of input pins 12/07 (DSF2) and 12/08 (DSF3) completed - Chapter 4.1.13. and 4.2.13. (Group 13 – Inputs): Value 1 of parameter 13/38 (Input DSF3) changed to “Engine Crank Inhibit” - Chapter 7.14.3. (Inducement Strategy for Tier4i Non-Road Engines): “Diesel Exhaust Fluid (DEF) level warning” corrected to “Diesel Exhaust Fluid (DEF) warning”, J1939 CAN message AUXIO1, signal in Byte 4, Bits 4,3 corrected to Status NOx Torque Limitation - Chapter 8.2. (Binary Values): Binary value 9/4 changed from “engine hood (bus)” to “DSF3 / Engine Crank Inhibit” - Chapter 11.2. (Applicable versions of the J1939 standard) and 11.4. (Overview of J1939 CAN messages) and 11.5. (CAN message details of miscellaneous messages): Message and/or signal descriptions adapted to newer versions of J1939 standard for messages TSC1, ERC1, ETC1, EEC2, EEC1, ETC7, AT1T1I, ACC1, EC1, PTO, CCVS1, LFE1, IC1, DD - Chapter 11.5.9. (ESS – Proprietary A (Engine Start/Stop)): Signal and value description of Engine Start/Stop Engine Shutdown Override corrected - Chapter 11.5.27. (AUXIO1 – Auxiliary Input/Output Status 1): Signal in Byte 4, Bits 2,1 changed to “Status Inducement” and Bits 4,3 changed to “Status NOx Torque Limitation” as described in chapter 7.14.3. (Inducement Strategy for Tier4i Non-Road Engines)

2011-06-22	10.00b	• Chapters 3.3.1 (21 pin Connector), 3.4.6. (Driver outputs (A)), 7.6.8. (former "Grid Heater indicator lamp", now "Wait to Start Lamp"), 7.12. (Cold start with grid heater), 10.1. (Routines for ADM3, Diagnosis version 210), 11.5.27. (AUXIO1 – Auxiliary Input/Output Status 1): "lamp Grid heater" or "Grid Heater Indicator" renamed to "Wait to Start Lamp", description of operating data range in J1939 CAN message AUXIO1 adapted • Chapter 3.3.2. (18 pin Connector): Override feature added for pin 18/13 (MABSCH_SP) • Chapters 3.3.2. (18 pin Connector), 3.3.4. (15 pin Connector), 4.1.2. and 4.2.2. (Group 02 – Veh Par I / Relay 2), 4.1.14. and 4.2.14. (Group 14 – Relay 3/4, AdBlue, NOx, misc.), 4.1.16. and 4.2.16. (Group 16 – Relay 1 / Starter Lockout), 7.14.1. (AdBlue Tank), 7.14.3. (Inducement Strategy for Tier4i Non-Road Engines): "AdBlue Level Warning Lamp" changed to "AdBlue Level/DEF Lamp" • Chapters 3.3.2. (18 pin Connector), 3.3.4. (15 pin Connector), 4.1.2. and 4.2.2. (Group 02 – Veh Par I / Relay 2), 4.1.14. and 4.2.14. (Group 14 – Relay 3/4, AdBlue, NOx, misc.), 4.1.16. and 4.2.16. (Group 16 – Relay 1 / Starter Lockout), 7.14.2. (NOx Torque Reduction Euro 4/5 / Euromot 3b Standard), 7.14.3. (Inducement Strategy for Tier4i Non-Road Engines): "NOx Lamp" and "TLA/NOx Lamp" changed to "LIM/NOx Lamp" • Chapters 4.1.8. and 4.2.8. (Group 08 – Vehicle Speed Sensor): Description of parameter 08/02 refined for password restricted parameter values, value descriptions for J1939 sources unified • Chapter 5.2. (Installation): Daimler part numbers for plugs and matching pins connected to the ADM3 added • Chapter 7.6.9. (Air filter indicator lamp): Remark, that the diagnostic routine for air filter differential pressure is done only in certain operating states • Chapter 7.9.1. (Ambient Air Temperature): Pin number for HFG+ at input of Air filter sensor corrected in Fig. 19 • Chapter 7.14. (Emission related functions – NOx torque reduction): chapters 7.13.5. and 7.13.6. moved to chapter 7.14. • Chapter 7.14.2. (NOx Torque Reduction Euro 4/5 / Euromot 3b Standard): chapter renamed, Euromot 3b standard included, weight classes for torque reduction listed • Chapter 7.14.3. (Inducement Strategy for Tier4i Non-Road Engines): Parameter settings for relay outputs corrected, value description of CAN signals in J1939 message Auxiliary Input/Output Status changed, ramping down of maximum engine speed in parallel to ramping down of maximum engine torque, figure for Tier 4i torque reduction as a function of time added, figure for Tier 4i engine speed reduction as a function of time added • Chapter 8.1. (Analog Values): Value descriptions of analog value 18 (Coolant level) changed, analog value 45 (Max.Retarder Torque) made signed • Chapters 9.1. (Fault codes diagnosis version 210) and 9.2. (Fault codes listed by K-line code): ADM3 fault codes SPN/FMI and K-line corrected by swapping inputs GAS1 and GAS2
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2011-04-08	10.00	- Chapters 4. (Parameters), 7. (Application), 8. (Actual Values), 9. (Fault Codes), 10. (Routines for ADM3): Diagnosis version changed to 210 - Chapters 4.1.1. and 4.2.1. (Group 1 – CAN Configuration), 11.3. (Default source addresses), 11.4.1. (Miscellaneous messages), 11.5.13. (ERC1 – Electronic Retarder Controller 1 (receive)), 11.5.46. (DD – Dash Display): Default value of parameter 01/08 (SAE J1939 Source Address Transmission Retarder) changed to 0, new parameter 01/19 SAE J1939 Source Address DDChapters 4.1.1. and 4.2.1. (Group 1 – CAN Configuration): Name of parameter 01/14 changed to SAE J1939 Source Address AMB, Chapters 4.1.10. and 4.2.10. (Group 10 – Engine Brake): Parameters 10/25 to 10/37 are changeable noChapters 4.1.11. and 4.2.11. (Group 11 – Accelerator Pedal): Parameter 11/25 (Accelerator Pedal Lock with Zero-Crossing) addedChapters 4.1.13. and 4.2.13. (Group 13 – Inputs): Parameter 13/39 (Configuration Ambient Temperature) addedChapters 4.1.14. and 4.2.14. (Group 14 – Relay 3/4, AdBlue, NOx, misc.): Min and Max values of parameters 14/23 and 14/24 changed, parameter 14/75 (NOx torque reduction Tier4i) addedChapters 4.1.17. and 4.2.17. (Group 17 – Idle/PTO Shutdown): Parameter 17/03 extended by values 4 to 6 for PTO Shutdown with Droop, parameters 17/10 (max. Speed for Droop Shutdown) and 17/11 (min. Torque for Droop Shutdown) addedChapters 4.1.22. and 4.2.22. (Group 22 – TSC1 Limiter Governor (N max)): Default value of parameter 22/01 (Gov# TSC1 Transmission) changed to 0Chapter 7.9. (Input Signals) and 7.9.1. (Ambient Temperature) addedChapter 7.13.4. (Accelerator pedal interlock): New accelerator pedal interlock functionality with zero-crossingChapter 7.13.6. (Emission related functions – NOx torque reduction): Previous text moved into subchapter 7.13.6.1. (Euro 4/ Standard) and chapter 7.13.6.2. (Inducement Strategy for Tier4i Non-Road Engines) addedChapter 8.1. (Analog values): Analog values 60 (Actual Torque Limiter Cutoff Factor) and 61 (Total Idle Hours) added - Chapter 8.2. (Binary Values): Binary value 11/3 (Tier4i System) added - Chapter 9.1. (Fault Codes): Four new failure codes (SPN/FMI 609/14, 1761/18, 4213/7, 3363/14) added due to update of MR2 diagnosis to version V23, SPN values 651 to 656 which were marked with footnote to be not SAE J1939 compliant in software 9.00 are referred now as SPN values 1268 to 1273 respectively, Fault codes SPN/FMI 171/3 (Ambient Air Temperature/Voltage too High) and 171/4 (Ambient Air Temperature/Voltage too Low) added - Chapter 10.1. (Routines for ADM3, Diagnosis version 210): Routines 32 (Delete Trip Data) and 33 (Total Idle Hours) and 34 (Engine Total Fuel Consumption) addedChapters 11.4.1. (Miscellaneous messages) and 11.5.32. (EC1 – Engine Configuration 1): Repetition rate of CAN message Engine Configuration 1 extended by sending on change of torque/speed points of more than 10% since last transmissionChapters 11.4.2. (Diagnostic messages) and 11.6.6. (DM4 – Diagnostic Message 4, Freeze Frame Parameters): SAE J1939 CAN message DM4 is supported now
2011-04-08	9.01d	Chapter 3.1. (Images of the vehicle control adaption module ADM3): Fig. 4 (View of connector side of ADM3) replaced, dimensioning of ADM3 unit added to Fig. 3 and Fig. 4Chapter 7.12.4. (Accelerator pedal interlock): Correction of contentsChapters 11.3. (Default source addresses) and 11.5.1. (TSC1 – Torque/Speed Control 1 (to Engine)) and 11.5.2. (TSC1 – Torque/Speed Control 1 (to Retarder)) and 11.5.9. (ESS – Proprietary A (Engine Start/Stop)) and 11.5.18. (EEC2 – Electronic Engine Controller 2 (receive)): Default source address of ADR3 corrected to 0xE7 corresponding to default value of parameter 01/03
2011-03-03	9.01c	Chapters 4.1.13. and 4.2.13. (Group 13 – Inputs): Values of parameter 13/03 (Enable Air Filter Sensor Input) restricted to 0 (disabled) and 1 (enabled)

2011-02-07	9.01b	- Chapters 4.1.2. and 4.2.2. (Group 02 – Veh Par I / Relay 2): Group name changed to Veh Par I / Relay 2 Hint for parameter 02/03 (Configuration Relay 2 / Grid Heater) added - Chapters 4.1.3. and 4.2.3. (Group 03 – Common Limiters): Minimum value of parameter 03/03 (Maximum Road Speed (legal)) changed to 40km/h Parameters 03/16 (Road Speed Limiter Special Parameter), 03/17 (Road Speed Limiter P-Factor), 03/18 (Road Speed Limiter Special Parameter) and 03/19 (Road Speed Limiter Special Parameter) changed to visible - Chapters 4.1.8. and 4.2.8. (Group 08 – Vehicle Speed Sensor): Remark about password protected values for parameter 08/02 (Vehicle Speed Sensor) added, Parameter 08/12 (Maximum valid C3-Voltage (Low-Level)) added as replacement to reserved entry - Chapters 4.1.10. and 4.2.10. (Group 10 – Engine Brake): Parameters 10/25 to 10/37 changed to visible - Chapters 4.1.11. and 4.2.11. (Group 11 – Accelerator Pedal): Hint for parameter 11/08 (PWM Pedal Kickdown Switch Off Threshold) added - Chapters 4.1.13. and 4.2.13. (Group 13 – Inputs): Description for parameter values 13/03 (Enable Air Filter Sensor Input) changed, Parameters 13/21 to 13/36 not changeable any more - Chapters 4.1.14. and 4.2.14. (Group 14 – Relay 3/4, AdBlue, NOx, misc.): Maximum value of parameter 14/32 (NOx torque reduction) changed to 100%, Description for parameters 14/72 (Lower faulty Resistance) and 14/73 (Upper faulty Resistance) added - Chapters 4.1.16. and 4.2.16. (Group 16 – Relay 1 / Starter Lockout): Value 2 of parameter 16/01 (Output Relay 1/ Starter Lockout) changed to acc. pedal kickdown - Chapters 4.1.24. and 4.2.24. (Group 24 – Vehicle Parameters II): Parameters 24/06 (Power Rating Selection), 24/07 (Power rating with Cruise Control) and 24/08 to 24/44 (Power Rating Curves 1 to 3, Values #0 to #15) made visible and changeable - Chapters 4.1.26. and 4.2.26. (Group 26 – VIN Vehicle Identification Number): All parameters 26/01 to 26/06 in this group are changed to not visible and not changeable - Chapter 9.1. (Fault codes diagnosis version 209): Fault Code table extended by column with MR2-Codes for respective failures - Chapter 9.1. (Fault codes diagnosis version 209) and 9.2. (Fault codes listed by K-line code): Fault description of fault 13901 (SPN 730 / FMI 1) and 13902 (SPN 730 / FMI 2) changed to Relay Permanently Closed (Grid Heater) and Relay Permanently Open (Grid Heater) respectively - Chapter 10. (Routines for ADM3, Diagnosis version 209) : Routine names changed for Nr. 6 (activate grid heater lamp), Nr. 8 (activate relay 1), Nr. 9 (activate relay 2), Nr. 10 (activate relay 3), Nr. 11 (activate relay 4) and Nr. 13 (activate MBR_KD), Routine Nr. 29 (TB-Ramp) added, Routine Nr. 30 (Torque Threshold) added Routine Nr. 31 (Maximum Road Speed (legal)) added
2009-12-18	9.01	- Chapter 9.1. (Fault codes diagnosis version 209) and 9.2. (Fault codes listed by K-line code): New fault codes SPN 1136 FMI 0 and FMI 1 (Engine ECU Temperature too High / too Low) from PLD/MR2 added, FMI values 5 and 6 for path 33 (Output Relay 1) changed to 3 and 4 respectively - Chapter 9.1. (Fault codes diagnosis version 209) and 9.2. (Fault codes listed by K-line code): The following changes only apply to engines which are equipped with a SCR system Proprietary Fault Code SPN 520229 FMI 7 changed to standard compliant SPN 4794 FMI 14, SPN 520261 FMI 12 changed to SPN 4332 FMI 12, FMI values for SPNs 1231, 3220 and 171 from PLD/MR2 changed from 8 to 9, FMI values for SPN 520262 from PLD/MR2 changed from 12 to 14 (Mannheim-Function and Würth-Function now have the same FMI), - Chapter 11.6.3. (DM1 – Diagnostic Message 1, Active Diagnostic Trouble Codes (send)): Repetition rate of J1939 CAN message DM1 adapted to sending also when there are no active faults

2009-12-18	9.00	- Chapters 4.1.1. and 4.2.1. (Group 01 – CAN Configuration): Parameters 01/17 (J1939 Timeout Detection) and 01/18 (SPN Conversion Method) added - Chapters 4.1.3. (Group 03 – Common Limiters): Maximum value of parameter 03/03 (Maximum Road Speed (legal)) changed - Chapters 4.1.13. and 4.2.13. (Group 13 – Inputs): Description of parameter value 0 of parameter 13/13 changed to disabled or stalk switch, Parameter values of parameter 13/37 (Input DSF2) extended by 2 (Enable TSC1 from SA3), 3 (Quit Signal for CC with Stalk Switch) and 4 (Power Rating with DSF2), default value changed from 1 to 0 - Chapters 4.1.18. and 4.2.18. (Group 18 – Engine Protection): Parameter 18/13 (Oil Level Lamp Configuration) added - Chapters 4.1.20. and 4.2.20. (Group 20 – Remote Accelerator Pedal): New parameter 20/07 (HFG Diagnostics Only With Active Pin "FG-Wahl") - Chapter 7.1.2.2. (Cruise control when using a stalk switch) added - Chapter 7.5.6. (Engine brake operation with a stalk switch) added - Chapter 7.6.6. (Oil level indicator lamp): The usage of the oil level lamp is extended with the help of the new parameter 18/13 (Oil Level Lamp Configuration) for indicating low oil level and engine overspeed as well - Chapter 7.12.8. (Securing of TSC1 message from third source) added - Chapter 7.12.9. (Power Rating with DSF2) added - Chapter 7.13.2. (J1939 Diagnosis) added - Chapter 8.1. (Analog Values): Analog Value "Ambient Air Temperature" added - Chapter 9. (Fault codes): Description of fault locations brought into accordance with newly integrated errors of J1939-signals - Chapter 9.1. (Fault codes diagnosis version 209): Errors broadcasted from the PLD/MR2 newly included because of possibility to be read out via J1939 - Chapter 9.1. (Fault codes diagnosis version 209) and 9.2. (Fault codes listed by K-line code): Change of diagnosis version from 207 to 209, Errors for J1939-signals (FMI 9: Signal Not Available, FMI 19: Received Network Data In Error) as well as messages (FMI 9: Abnormal Update Rate) added, Failure detection of Decompression Valve MBR_KD (ADM3 path 0) and Intake Air Heater (now residing on ADM3 path 40, SPN 729) separated, SPN of path 0 (Decompression Valve MBR_KD) changed to 1072, SPN of path 46 (Exhaust Brake Valve MBR_BK) changed to 1074, Fault codes 11814 (SPN 599, FMI 14), 11914 (SPN 601, FMI 14), 12002 (SPN 609, FMI 2), 19214 (SPN 1633, FMI 14) added - Chapter 9.2. (Fault codes listed by K-line code) added for reverse search of failure codes - Chapters 11.4.1. (Miscellaneous messages) and 11.5.3. (RESET – Reset): Receive message RESET for deleting of trip data over J1939 added - Chapters 11.4.2 (Diagnostic messages), 11.6.1. (Diagnostic Message 13 – Stop Start Broadcast), 11.6.5. (Diagnostic Message 3 – Diagnostic Data Clear/Reset Of Previously Active DTCs), 11.6.6. (Diagnostic Message 4 – Freeze Frame Parameters) and 11.6.7. (Diagnostic Message 11 – Diagnostic Data Clear/Reset For Active DTCs) added
2009-12-16	7.03j	- Chapter 3.2. (Wiring diagram of ADM3) added - Chapter 3.3.3. (12 pin Connector): Description of pin 12/07 corrected to configurable input DSF2 - Chapters 3.3.2. (18 pin Connector) and 3.3.3. (12 pin Connector) and 3.3.4. (15 pin Connector): Parameter numbers noted for relays 1 to 4, DSF0, DSF1, DSF2, IWA - Chapter 4.1. (List of Parameters): Correction of default values of parameters 02/05, 11/10, 14/01, 14/10, 14/21, 16/01, 19/09, 19/10, 19/13 and 19/14 - Chapter 4.1. (List of Parameters): Completion of units in parameter tables - Chapter 4.1. (List of Parameters): Correction of maximum values of parameters 13/02, 13/07 and 14/74 - Chapter 4.1.5 (Group 05 – Limiters LIM0 and LIM1) and 4.1.6. (Group 06 – Limiters AC/LIM2): Minimum values of parameters 05/04, 05/08 and 06/06 changed to -2000Nm - Chapter 4.1.22 (Group 22 – TSC1 Limiter Governor (N max)): Maximum value of parameter 22/01 changed to 4, Parameters 22/01 to 22/04 are visible now - Chapter 7.12.7. (TSC1 Speed Control and Speed Limitation) added - Chapter 8.2. (Binary Values): Binary value 11/2 changed to DSF2 Status - Chapter 9.1. (Fault codes diagnosis version 207): Fault codes 10202, 10900, 12312, 15002 deleted because they are not supported by software - All chapters: Uniform notation PLD/MR2 for engine controller
2009-04-20	7.03i	- Chapter 4.1.22. (Group 22 – TSC1 Limiter Governor (n max)): Default value of parameter 22/20 corrected to 0 - Chapter 11.5.25. (EBC2 – Wheel Speed Information): Signal names of Bytes 5 to 8 corrected

2009-02-10	7.03h	- Chapter 7.3.1. (PWM accelerator pedal): Correction of abstract "Various PWM accelerator pedal types" - Chapter 7.5. (Engine Brake (Retarder)): Parameter 02/13 corrected to 10/13 (Engine Brake Configuration) - Chapter 7.5.3.4. (Exhaust flap and decompression valve at one valve): X 15/10 changed to X 15/06 in Fig. 12 - Chapter 7.5.3. (Engine brake configuration): Reference to PLD parameters 06/01 and 06/02 extended by reference to MR2 parameters depending on diagnosis version in subchapters 7.5.3.1., 7.5.3.2., 7.5.3.3., 7.5.3.4., 7.5.3.5.1., 7.5.3.5.2. and 7.5.3.5.3., "analog valve" changed to "proportional valve"
2009-01-14	7.03g	- Chapters 3.2.1. (21 pin Connector): Description of pin 21/16 corrected - Chapters 3.2.2. (18 pin Connector), 4.1.14. and 4.2.14. (Relay 3/4, AdBlue, NOx, misc.): Parameter value 11 of parameter 14/10 (Configuration Relay 4 (IWK4)) changed to "Reserved" - Chapters 4.1.25 and 4.2.25 (XTempomat) and 4.1.28 and 4.2.28 (Real TimeClock) added - Chapters 4.1.2. and 4.2.2. (Group 2 – Vehicle Parameters I): Parameter value 2 of parameter 02/09 described more precisely - Chapter 4.1.11. (Group 11 – Accelerator Pedal): Default value of parameter 11/08 (PWM Pedal Kickdown Switch Off Threshold) changed to 11% - Chapters 4.1.19. and 4.2.19. (Group 19 – Fan): Description of parameters 13 (DSF0 Fan) and 14 (DSF1 Fan) added - Chapters 4.1.22 (Group 22 – TSC1 Limiter Governor (n max)): Description of not visible parameters 1-8 and 16-18 eliminated - Chapter 7.3.1. (PWM accelerator pedal): Description of parameter issues caused by new PWM pedal type added - Chapters 7.12.2. (Conventional Retarder) and 7.12.3. (Automatic Fan): Correction of parameter numbering of parameters in group 19 (Fan) and 13 (Inputs) - Chapter 7.12.3. (Automatic Fan): Correction of parameter numbering in Fig. 18 - Release of an updated version 6.06d of this manual containing these and other corrections from the release 7 manuals, for detailed description of the changes compare manual 6.06d
2008-07-16	7.03f	- Chapter 4.1.11. (Group 11 – Accelerator Pedal): Default value of parameter 11/09 corrected to 0 - Release of an updated version 6.06c of this manual containing these and other corrections from the release 7 manuals, for detailed description of the changes compare manual 6.06c
2008-02-07	7.03e	- Chapter 11.5.3. (ACK / NACK – Acknowledgment): Default Sender Address from Retarder corrected to 0x0F - Chapter 11.6.3. (DM2 – Diagnostic Message 2, Previously Active Diagnostic Trouble Codes): Identifier corrected to 0x18FECBxx (xx = Sender Address) - Release of an updated version 6.06b of this manual containing these and other corrections from the release 7 manuals, for detailed description of the changes compare manual 6.06b
2008-01-18	7.03d	- Chapter 3.2.4. (15 pin Connector): Description of PIN 15/01 corrected - Chapter 4.1.3. (Group 03 – Common Limiters): Max value of parameter 03/03 corrected
2007-10-17	7.03c	- Chapter 3.2.2. (18 pin Connector): Description of PIN 18/01 updated - Chapter 3.2.3. (12 pin Connector): Description of PIN 12/05, 12/09, 12/10 updated - Chapter 3.2.4. (15 pin Connector): Description of PIN 15/09, 15/11, 15/12 updated - Chapter 4.1.2. (Group 02 – Vehicle Parameters I): Max value of number 03 (Configuration Relay 2 / Grid heater) incremented to 8 - Chapter 4.1.14. (Group 14 – Relay 3/4, AdBlue, NOx, misc.): Max value of number 01 (Configuration Relay 3 (IWK3)) incremented to 13 Max value of number 10 (Configuration Relay 4 (IWK4)) incremented to 14 - Chapter 4.1.16. (Group 16 – Relay 1 / Starter Lockout): Max value of number 01 (Output Relay 1/Starter Lockout) incremented to 9 - Chapter 4.2.14. (Group 14 – Relay 3/4, AdBlue, NOx, misc.): Parameter values for number 01 and 10 updated - Chapter 8.1. (Analog Values): Description of value 18 changed from "%" to 0 = Switch defective 1 = Warning 2 = Pre warning 3 = Coolant level normal - All Chapters: changed DaimlerChrysler to Daimler
2007-10-15	7.03b	Chapter 9. (Fault codes): Fault 639/2 is renamed from "CAN identifier ETC#1 or ACC#1 is missing" to "At least one J1939 message is missing"

2007-09-13	7.03	- Chapter 4. (Parameters): Change of order of parameters 01/13, 01/14, 01/16 in group 01: "Send Free Running Telegram on K line" from 01/13 to 01/16, "SAE J1939 Source Address ACC" from 01/14 to 01/13, "SAE J1939 Source Address Ambient Temperature" from 01/16 to 01/14; - Chapter 4. (Parameters): Extension of parameter values for relays 1..4 for NOx lamp, Relay 1 Par. 16/01 case 9, Relay 2 Par. 02/03 case 8, Relay 3 Par. 14/01 case 13, Relay 4 Par. 14/10 case 14; - Chapter 4. (Parameters): Move of parameter "Fan Type" from 14/21 to 19/01, therefore change of numbering of subsequent parameters in both groups; - Chapter 4. (Parameters): Parameter 14/33 "freebyte 2" changed to 14/32 "NOx torque reduction"; - Chapter 4. (Parameters): Group 14 is renamed from "Relay 3/4" to "Relay 3/4, AdBlue, NOx, misc.", Group 19 is renamed from "Automatic Fan Activation" to "Fan"; - Chapter 4. (Parameters): Change of default values of parameters 18/01 "Engine Protection Shutdown on Coolant Temperature" from 1 to 0 and 18/03 "Engine Protection Shutdown on Oil Pressure" from 1 to 0; - Chapter 4. (Parameters): Change of default value of parameter 19/01 "Fan Type" from 0 to 251; - Chapter 11. (CAN messages according to SAE J1939): CAN message "Auxiliary Input/Output Status 1" extended by the status information of the NOx torque limitation (Byte 4, Bits 4,3); - Chapter 11. (CAN messages according to SAE J1939): Length of CAN message "Engine Configuration 1" changed from 29 to 34 in accordance to newer versions of the J1939 standard; - Chapter 11. (CAN messages according to SAE J1939): Integration of malfunction indicator lamp in message "Diagnostic Message 1" (sending), support of flashing of amber warning lamp, red stop lamp and malfunction indicator lamp; - Chapter 11. (CAN messages according to SAE J1939): Support of CAN message "Diagnostic Message 2".
2007-09-13	6.06	- Complete redesign of the operating manual; - Version number of the operating manual corresponds with the delivered ADM3 software version; - Chapter 11. (CAN messages according to SAE J1939) rewritten because of a lot of new supported J1939 CAN messages since software version 4.23 of the last operating manual release; - Significant changes in chapter 4. (Parameters) and a few changes in chapter 9. (Fault codes) and 10. (Routines for ADM3).

Table of Abbreviations

Abbreviation	Meaning
ABS	Anti-lock braking system
ADM3	Adaption module as vehicle control, new version, also denoted as ADM3-FR in this manual
ADR	PTO speed control
BK	Engine retarder flap, also MBR-BK
C3/B7	Speed signal C3/B7
CAN	Control Area Network
CC+	Cruise Control Resume and Acceleration
CC-	Cruise Control Set and Decelerate
CC_EIN	Cruise Control, Cruise control on-off switch
EEPROM	Electrical erasable and programmable read only memory
EMV	EMC/electromagnetic compatibility
EWG	European economic communities in the European Community, precursor of the EU
FFG	The foot throttle actuator is the accelerator pedal
FMR	Vehicle control for Mercedes-Benz commercial vehicles type Actros or type Atego
FSBE	Input for the switching state of the parking brake
HFG	Remote pedal
Highside Schalter	Switch (switched to battery voltage)
IWA	Actual value output
K-Leitung	Serial communication- and diagnosis line
KD	Constantly open valve, also MBR-KD
Lowside Schalter	Switch (switched to ground)
MBR	Engine brake
MCAN	Engine CAN data bus between ADM3 and PLD/MR2
Minidiag 2	Diagnosis- and configuration unit for the ADM3
MR2	Engine control unit, successor of PLD unit
NE	Input for transmission position Neutral
PLD/MR2	Engine control type PLD (for the injection principle pump-line-nozzle)
PTO	Power Take Off
PWM	Pulse width modulation
SAE J1939	CAN data bus according to standard SAE J1939

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